Lecture 5 Homework

Programming

None

Exercises

1. An RSA cipher is set up with $\text{Key}_{\text{pub}} = (210757, 3)$ where 210757 is the modulus and 3 is the exponent. Encrypt $\text{Plaintext} = 12345$.
2. An RSA cipher is set up with $\text{Key}_{\text{pub}} = (15943\ , 3)$, where 15943 is the modulus and 3 is the exponent. Find the decryption key.
3. An RSA cipher is set up with $\text{Key}_{\text{pub}} = (12091\ , 3)$ modulus 12091 and the encryption key is 3. Decrypt $\text{Ciphertext} = 9812$.

Suggested Reading

- Garrett – Sections 10.2,
- Smart – Sections 7.3, 11.3, , 17.3.2, 18.2.1
- Forouzan – Sections 10.2, 13.5